

Estimated Annual Agricultural Pesticide Use Marion County, Oregon 2014-2018

24 July 2026

Summary

Marion County is Oregon’s largest and most diversified agricultural county by production value, anchoring the northern Willamette Valley. It is nationally recognized for nursery stock, berries (particularly blackberries, marionberries — its namesake cultivar — and other caneberries), hazelnuts, and vegetable and grass seed crops, playing a role in Oregon agriculture comparable to Yakima County’s for Washington’s tree-fruit and hop industries in the companion Washington chapter of this repository.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Marion County, Oregon** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound’s chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” [-Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OP	Organophosphorous compound
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Marion County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	BACILLUS AMYLOLIQUEFACIEN	MIC	54366	18153	18153
2	CHLOROPICRIN	OC	43978	14861	9746
3	GLYPHOSATE	OTH	38334	21293	17969
4	METAM	DTC	31494	50927	35919
5	CALCIUM POLYSULFIDE	INO	23090	21734	17231
6	SULFUR	INO	22997	15893	13820
7	PETROLEUM OIL	OTH	16264	11804	12912
8	GLUFOSINATE	OTH	9766	6311	5087
9	ZIRAM	DTC	7935	2808	1795
10	PARAQUAT	DIP	7149	5729	4841
11	COPPER SULFATE	INO	7101	7677	6389
12	EPTC	TC	6583	8239	8826
13	ETHOPROPHOS	OP	5036	5413	5532
14	METRIBUZIN	TRZ	4428	2515	1674
15	METOLACHLOR & METOLACHLOR-S	OTH	4078	4695	5085
16	METOLACHLOR-S	OTH	4078	4695	5058
17	FLUFENACET	OTH	3962	2472	1744
18	BURKHOLDERIA SPP	MIC	3738	3797	3797
19	ATRAZINE	TRZ	3489	3889	4022
20	PROPICONAZOLE	OTH	3444	2008	1845
21	CHLOROTHALONIL	OTH	3221	9550	7592
22	CHLORPYRIFOS	OP	3043	5141	4823
23	AZOXYSTROBIN	OTH	2891	1436	1308
24	SIMAZINE	TRZ	2605	1856	2377
25	2,4-D	CPX	2548	2754	3248
26	THIOPHANATE-METHYL	OTH	2464	3917	3545
27	DIURON	OTH	2456	2711	3292
28	CAPTAN	OTH	2282	2879	3562
29	MCPA	CPX	2016	1790	1379
30	BENTAZONE	OTH	1918	2183	2173
31	COPPER	INO	1895	2176	2143
32	IPRODIONE	OTH	1736	2589	2940
33	DCPA	OTH	1567	2580	2357
34	MANCOZEB	DTC	1362	6234	5550
35	PENDIMETHALIN	OTH	1312	2141	1916
36	COPPER HYDROXIDE	INO	1086	1176	1122
37	DIMETHENAMID & DIMETHENAMID-P	OTH	1028	1095	1250
38	DIMETHENAMID-P	OTH	1028	1095	1250
39	DIMETHOATE	OP	900	606	580
40	BROMOXYNIL	OTH	672	547	595
41	CLOPYRALID	OTH	637	400	272
42	FLUROXYPYR	OTH	637	313	232
43	FENHEXAMID	OTH	609	480	680
44	DICAMBA	OTH	587	497	447
45	BOSCALID	OTH	580	502	671

Table 2: Top 80 EPest-high Estimates, Marion County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
46	TRIFLURALIN	OTH	556	680	656
47	MALATHION	OP	534	845	1457
48	TRIFLOXYSTROBIN	OTH	501	600	582
49	CYHALOTHRIN-LAMBDA	PYR	476	304	214
50	FLUMIOXAZIN	OTH	448	274	273
51	COPPER OXYCHLORIDE	INO	421	311	246
52	CYPRODINIL	OTH	402	369	458
53	ESFENVALERATE	PYR	394	348	367
54	PINOXADEN	OTH	372	135	110
55	SAFLUFENACIL	OTH	366	187	125
56	PYRACLOSTROBIN	OTH	338	394	458
57	CLETHODIM	OTH	284	485	371
58	FLUDIOXONIL	OTH	266	242	289
59	BACILLUS THURINGIENSIS	MIC	261	353	232
60	DICHLOROPROPENE	OC	260	1230	5098
61	BENZOVINDIFLUPYR	OTH	254	254	254
62	MESOTRIONE	OTH	245	266	164
63	OXAMYL	CB	242	82	211
64	CARFENTRAZONE-ETHYL	OTH	236	152	122
65	METHOMYL	CB	224	1740	1464
66	TRIFLUMIZOLE	OTH	202	186	238
67	BIFENTHRIN	PYR	202	185	186
68	CYFLUTHRIN	PYR	200	90	81
69	PENTHIOPYRAD	OTH	198	152	149
70	INDAZIFLAM	OTH	196	171	124
71	PYROXASULFONE	OTH	195	109	97
72	IMIDACLOPRID	OTH	194	130	142
73	THIFENSULFURON	OTH	189	87	76
74	OXYFLUORFEN	OTH	152	637	507
75	MCPB	CPX	142	90	63
76	FOMESAFEN	OTH	127	127	106
77	METAM POTASSIUM	DTC	126	96	963
78	SETHOXYDIM	OTH	119	251	247
79	TRIBENURON METHYL	OTH	116	59	54
80	ETHOFUMESATE	OTH	115	973	743

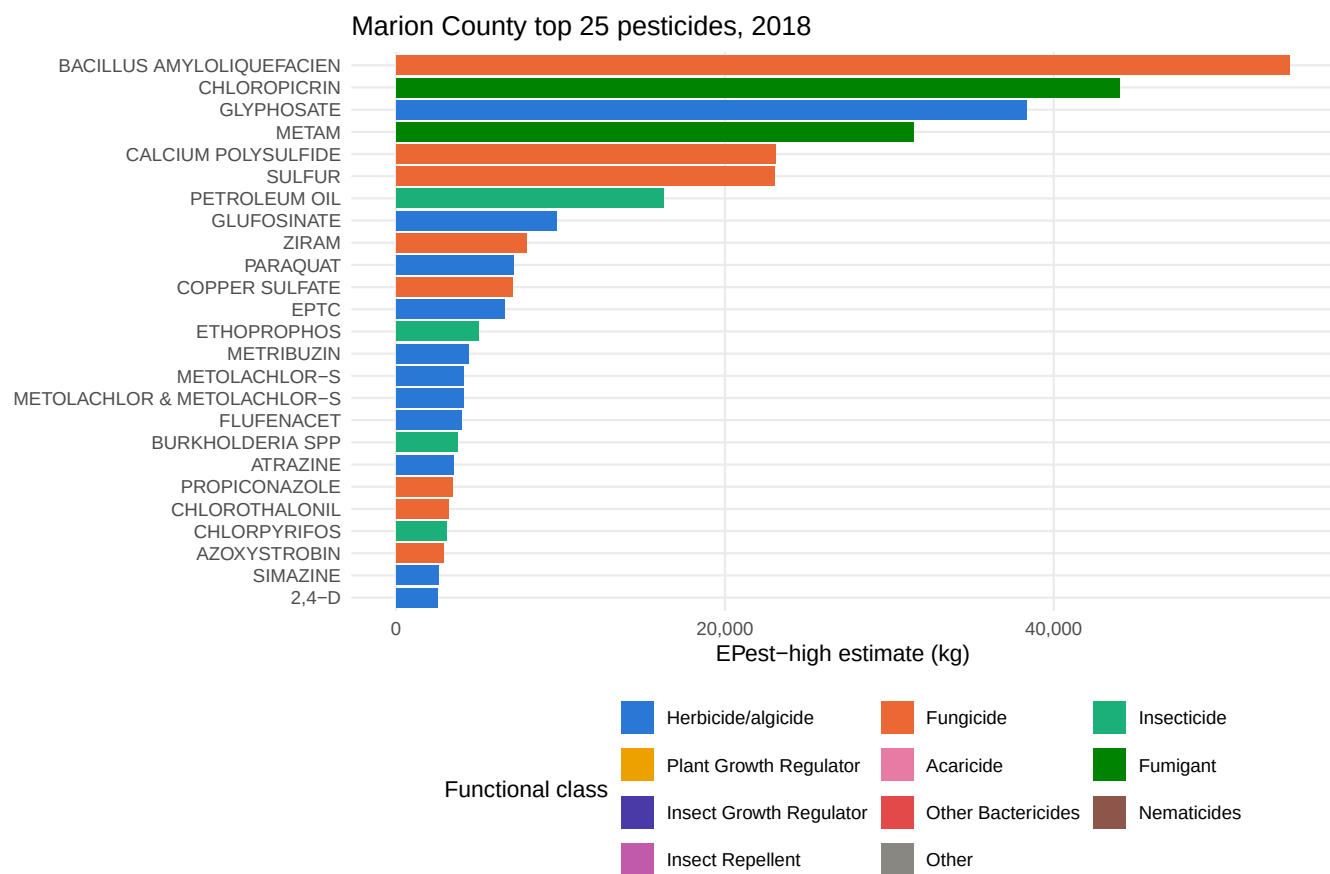


Figure 1: Top 25 compounds in Marion County by EPest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Marion County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	BACILLUS AMYLOLIQUEFACIEN	0	54366
2	CHLOROPICRIN	0	43978
3	GLYPHOSATE	37860	38334
4	METAM	0	31494
5	CALCIUM POLYSULFIDE	23090	23090
6	SULFUR	22997	22997
7	PETROLEUM OIL	16264	16264
8	GLUFOSINATE	9766	9766
9	ZIRAM	7935	7935
10	PARAQUAT	5418	7149
11	COPPER SULFATE	7101	7101
12	EPTC	6580	6583
13	ETHOPROPHOS	5036	5036
14	METRIBUZIN	4425	4428
15	METOLACHLOR & METOLACHLOR-S	4072	4078
16	METOLACHLOR-S	4072	4078
17	FLUFENACET	3962	3962
18	BURKHOLDERIA SPP	0	3738
19	ATRAZINE	3489	3489
20	PROPICONAZOLE	3444	3444
21	CHLOROTHALONIL	3204	3221
22	CHLORPYRIFOS	2942	3043
23	AZOXYSTROBIN	2891	2891
24	SIMAZINE	2605	2605
25	2,4-D	1191	2548
26	THIOPHANATE-METHYL	2396	2464
27	DIURON	2428	2456
28	CAPTAN	2282	2282
29	MCPA	2016	2016
30	BENTAZONE	1918	1918
31	COPPER	1895	1895
32	IPRODIONE	1736	1736
33	DCPA	0	1567
34	MANCOZEB	1354	1362
35	PENDIMETHALIN	1202	1312
36	COPPER HYDROXIDE	662	1086
37	DIMETHENAMID & DIMETHENAMID-P	789	1028
38	DIMETHENAMID-P	789	1028
39	DIMETHOATE	881	900
40	BROMOXYNIL	426	672
41	CLOPYRALID	637	637
42	FLUROXYPYR	637	637
43	FENHEXAMID	609	609
44	DICAMBA	469	587
45	BOSCALID	388	580
46	TRIFLURALIN	556	556
47	MALATHION	47	534
48	TRIFLOXYSTROBIN	501	501
49	CYHALOTHRIN-LAMBDA	466	476
50	FLUMIOXAZIN	448	448

Table 4: Top 50 Marion County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	0	50927
2	CALCIUM POLYSULFIDE	20934	21734
3	GLYPHOSATE	20463	21293
4	BACILLUS AMYLOLIQUEFACIEN	0	18153
5	SULFUR	15268	15893
6	CHLOROPICRIN	0	14861
7	PETROLEUM OIL	11804	11804
8	CHLOROTHALONIL	9544	9550
9	EPTC	8231	8239
10	COPPER SULFATE	7543	7677
11	GLUFOSINATE	6309	6311
12	MANCOZEB	6229	6234
13	PARAQUAT	4587	5729
14	ETHOPROPHOS	5413	5413
15	KAOLIN CLAY	643	5374
16	CHLORPYRIFOS	5063	5141
17	METOLACHLOR & METOLACHLOR-S	4690	4695
18	METOLACHLOR-S	4690	4695
19	CUPROUS OXIDE	4257	4257
20	THIOPHANATE-METHYL	3841	3917
21	ATRAZINE	3946	3889
22	BURKHOLDERIA SPP	0	3797
23	PHOSPHORIC ACID	3331	3361
24	CAPTAN	2879	2879
25	ZIRAM	2721	2808
26	2,4-D	1346	2754
27	DIURON	2324	2711
28	IPRODIONE	2589	2589
29	DCPA	0	2580
30	METRIBUZIN	2496	2515
31	FLUFENACET	2472	2472
32	BENTAZONE	2179	2183
33	COPPER	2084	2176
34	PENDIMETHALIN	1992	2141
35	DICHLORBENIL	2092	2092
36	PROPICONAZOLE	2008	2008
37	SIMAZINE	1853	1856
38	MCPA	1662	1790
39	METHOMYL	1739	1740
40	AZOXYSTROBIN	1417	1436
41	DICHLOROPROPENE	0	1230
42	COPPER HYDROXIDE	625	1176
43	DIMETHENAMID & DIMETHENAMID-P	997	1095
44	DIMETHENAMID-P	997	1095
45	ETHOFUMESATE	973	973
46	MALATHION	318	845
47	TRIFLURALIN	670	680
48	OXYFLUORFEN	606	637
49	DIMETHOATE	556	606
50	TRIFLOXYSTROBIN	598	600

Table 5: Top 50 Marion County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	0	35919
2	BACILLUS AMYLOLIQUEFACIEN	0	18153
3	GLYPHOSATE	16742	17969
4	CALCIUM POLYSULFIDE	16466	17231
5	SULFUR	12653	13820
6	PETROLEUM OIL	12469	12912
7	CHLOROPICRIN	0	9746
8	EPTC	8790	8826
9	CHLOROTHALONIL	7570	7592
10	COPPER SULFATE	6274	6389
11	MANCOZEB	5546	5550
12	ETHOPROPHOS	5532	5532
13	DICHLOROPROPENE	0	5098
14	GLUFOSINATE	4935	5087
15	METOLACHLOR & METOLACHLOR-S	4943	5085
16	METOLACHLOR-S	4916	5058
17	PARAQUAT	4070	4841
18	CHLORPYRIFOS	4742	4823
19	KAOLIN CLAY	1334	4614
20	ATRAZINE	4057	4022
21	BURKHOLDERIA SPP	0	3797
22	CAPTAN	3562	3562
23	THIOPHANATE-METHYL	3405	3545
24	DIURON	2987	3292
25	2,4-D	1775	3248
26	CUPROUS OXIDE	2451	3054
27	IPRODIONE	2940	2940
28	SIMAZINE	2352	2377
29	DCPA	0	2357
30	PHOSPHORIC ACID	2232	2247
31	BENTAZONE	2132	2173
32	COPPER	1965	2143
33	PENDIMETHALIN	1715	1916
34	PROPICONAZOLE	1833	1845
35	ZIRAM	1712	1795
36	FLUFENACET	1744	1744
37	METRIBUZIN	1633	1674
38	METHOMYL	1463	1464
39	MALATHION	1023	1457
40	MCPA	1291	1379
41	DICHLOBENIL	1326	1326
42	AZOXYSTROBIN	1287	1308
43	DIMETHENAMID & DIMETHENAMID-P	1157	1250
44	DIMETHENAMID-P	1157	1250
45	COPPER HYDROXIDE	684	1122
46	METAM POTASSIUM	0	963
47	ACEPHATE	811	811
48	NAPROPAMIDE	569	780
49	ORYZALIN	611	750
50	ETHOFUMESATE	743	743